

2

(a)

Ans//

$$C = \begin{bmatrix} 0.25 & 0.20 \\ 0.20 & 0.25 \end{bmatrix}$$

EIGEN VALUES:

$$\begin{vmatrix} 0.25 - \lambda & 0.2 \\ 0.2 & 0.25 - \lambda \end{vmatrix} = 0$$

$$\Rightarrow (\lambda - 0.25)^2 - 0.2^2 = 0$$

$$\Rightarrow (\lambda - 0.25 + 0.2)(\lambda - 0.25 - 0.2) = 0$$

$$\Rightarrow \lambda_1 = 0.45 \quad \lambda_2 = 0.05$$

EIGEN VECTORS:

For $\lambda_1 = 0.45$

$$\begin{bmatrix} -0.2 & 0.2 \\ 0.2 & -0.2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_1 = v_2$$

$$\therefore e_1 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

For $\lambda_2 = 0.05$

$$\begin{bmatrix} 0.2 & 0.2 \\ 0.2 & 0.2 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow v_1 = -v_2$$

$$\therefore e_2 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

 $\therefore \lambda_1 > \lambda_2$,the first principal component is $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$

$$\begin{aligned}
 \text{Percentage of variance explained by PC}_1 &= \frac{\lambda_1}{\lambda_1 + \lambda_2} \\
 &= \frac{0.45}{0.50} \\
 &= 90\% //
 \end{aligned}$$

To pr

PROJECTION

$$n \times x \in \mathbb{R}^{784}$$

Step (1) : Compute covariance matrix $C \in \mathbb{R}^{784 \times 784}$

Step 2 : Compute its eigen values, sort in decreasing order

$$u_1 > u_2 \dots > u_{784}$$

Step 3 : Retain only first k eigen values vectors corresponding to eigen values

$$U_k = \begin{bmatrix} | & | & & | \\ e_1 & e_2 & \dots & e_k \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{784 \times k}$$

Step 4: Center the input:

$$\bar{x} = x - \mu \quad [\mu \in \mathbb{R}^{784}]$$

Step 5: Projection,

$$z = U_k^T (x - \mu) \quad [z \in \mathbb{R}^k]$$

3.

(a) BRIEF SUMMARY:

The paper introduces a set of regression models built specifically for data where the dependent variable is ordinal, i.e. they have a logical order.

Instead of just assigning numerical scores to these categories which assumes the distance between each category is the same, it assumes that there is a latent variable that we can't measure directly. The model treats the target variable (discrete and ordered) as contiguous blocks / intervals along this hidden continuous scale. These blocks are separated by unknown boundary markers called "cut points" (θ_j). By this way, we analyze the data based purely on its order without forcing fake scores on it.

The paper specifically highlights two models that are useful for this: proportional odds model & proportional hazards model. Essentially these models help us measure a "stochastic ordering" i.e. if a specific factor shifts the response higher or lower, the model captures the shift consistently across all category boundaries.

The paper also applies the model to real world examples: (i) it shows how you can quantify exactly how much being a bacteria carrier increases the odds of having enlarged tonsils across multiple size categories; (ii) it also shows how the

PTO

severity of disturbed dreams smoothly decreases by a specific factor as boys age from 5 to 15. The model essentially summarizes this overall relationship to a single parameter that captures how the distribution shifts across groups making the results much more useful for quantitative descriptions.

LIKELIHOOD & ODDS RATIO:

Ordinal regression relies on cumulative probabilities rather than probability of falling into a certain class.

If $\pi_j(x) :=$ probability of being in category j , the cumulative probability is defined as:

$$\delta_j(x) = \pi_1(x) + \pi_2(x) + \dots + \pi_j(x)$$

The odds of being in category j or lower is written as $K_j(x)$.

$$K_j(x) = \frac{\delta_j(x)}{1 - \delta_j(x)}$$

In proportional odds model, this is expressed using linear logistic equation

$$\log[K_j(x)] = \theta_j - \beta^T x$$

$\therefore$ the odds ratio for two different data points x_1 & x_2 is

$$\frac{K_j(x_1)}{K_j(x_2)} = \exp\{\beta^T (x_2 - x_1)\}$$

which is completely independent of the specific category j being looked at

To calculate likelihood, we start with standard multinomial likelihood, $\pi_1^{n_1} \dots \pi_k^{n_k}$ and transform it into cumulative totals.

If $n_j := \text{count in category } j$

the cumulative sums are defined as:

$$R_1 = n_1$$

$$R_2 = n_1 + n_2 \text{ and so on}$$

$$R_k = n$$

The likelihood is factored into product of $k-1$ conditional probabilities. For instance, the first part is the probability that the first two cells split in ratio $R_1 : R_2 - R_1$,

$$\text{i.e. } \left\{ \left(\frac{r_1}{r_2} \right)^{R_1} \left(\frac{r_2 - r_1}{r_2} \right)^{R_2 - R_1} \right\} \Rightarrow \text{likelihood naturally takes in the cumulative ordered nature of the data.}$$

Multiclass classification handles likelihood & odds ratio differently $\because$ the categories do not have a strict order.

$$\therefore p(y=k|x) = \frac{\exp(w_k^T x)}{\sum_{i=1}^K \exp(w_i^T x)} \quad [\text{using the softmax function}]$$

For odds ratio,

$$\frac{p(y=k|x)}{p(y=j|x)} = \frac{\exp(w_k^T x)}{\exp(w_j^T x)} = \exp[(w_k - w_j)^T x]$$

Unlike this, in ordinal regression, because of the proportional odds assumption, it uses a single shared set of weights, β , to describe the shift.

DIFFERENCE FROM REGRESSION PROBLEMS

In standard regression problems, we want to predict a continuous, real-valued output $y \in \mathbb{R}$.

For linear regression

$$y = w^T x + \epsilon \quad \text{where } \epsilon \sim \mathcal{N}(0, \sigma^2)$$

polynomial regression,

$$y = \sum_{i=0}^M w_i x^i \quad \text{where } M := \text{order of the polynomial}$$

In both cases, we try to predict the exact numerical value.

Ordinal regression works differently. The target variable falls into set of discrete categories rather than being continuous. Because a std regression line can't be fit into qualitative bins, ordinal models apply a link function (like logit or probit). $\therefore$ instead of estimating continuous outputs directly, the ordinal model ~~settles~~ estimates unknown boundary cut points (θ_j) that divide the ordered categories on the hidden continuous scale.

3
(b)

Suppose the response variable Y has k ordered categories

$$\text{i.e. } Y = 1, 2, \dots, k$$

$$\text{Let } \pi_j(x) = P(Y=j|x)$$

Let the cumulative probability be defined as

$$y_j(x) = P(Y \leq j|x) = \sum_{r=1}^j \pi_r(x)$$

$$\therefore \text{Category probability, } \pi_j = y_j - y_{j-1} \quad \begin{array}{l} y_0 = 0 \\ y_k = 1 \end{array}$$

Proportional Odds Model:

$$\log \left[\frac{y_j(x)}{1 - y_j(x)} \right] = \theta_j - \beta^T x \quad j \in [k-1]$$

θ_j : category specific cut point

$\Rightarrow$ same β occurs for every j

β : regression coefficient

x : covariate vector

$$\therefore y_j = F(\theta_j - \beta^T x)$$

where F is the logistic function

$$F(z) = \frac{e^z}{1 + e^z}$$

The cut pts must satisfy

$$\theta_1 < \theta_2 < \dots < \theta_{k-1}$$

so that cumulative probability is

increasing

Suppose there are n observations & n_j fall into category j

$$\Rightarrow \sum_{j=1}^k n_j = n$$

$$\Rightarrow (n_1, n_2, \dots, n_k) \sim \text{Multinomial}(n; \pi_1, \dots, \pi_k)$$

$$\therefore \text{likelihood, } L = \frac{n!}{n_1! \dots n_k!} \left[\prod_{j=1}^k \pi_j^{n_j} \right]$$

$$L \propto \prod_{j=1}^k \pi_j^{n_j}$$

Taking logarithms,

$$\begin{aligned} l &= \sum_{j=1}^k n_j \log \pi_j \\ &= \sum_{j=1}^k [n_j \log (y_j - y_{j-1})] \end{aligned}$$

$$\text{Let } R_j = \sum_{i=1}^j n_i \quad \text{and}$$

$$z_j = \frac{R_j}{n} \quad [z_k = 1]$$

Consider $\frac{y_j}{y_{j+1}}$

$$\text{with } y_j = P(Y \leq j) \text{ \& } y_{j+1} = P(Y \leq j+1)$$

$$\Rightarrow \frac{y_j}{y_{j+1}} = P(Y \leq j | Y \leq j+1) \propto \frac{z_j}{z_{j+1}}$$

$\therefore$ each stage behaves like a binomial model

$$\text{Let } \phi_j = \log \left(\frac{y_j}{y_{j+1} - y_j} \right) \quad ; \quad \frac{y_j}{y_{j+1} - y_j} = \frac{y_j / y_{j+1}}{1 - y_j / y_{j+1}}$$

$$\Rightarrow \phi_j = \text{logit} \left(\frac{y_j}{y_{j+1}} \right) \Rightarrow \frac{y_j}{y_{j+1}} = \frac{e^{\phi_j}}{1 + e^{\phi_j}} = p_j //$$

$$\text{Let } g(\phi) = \log(1 + e^\phi)$$

$$\Rightarrow g'(\phi) = \frac{e^\phi}{1 + e^\phi} = p_j$$

$$g''(\phi) = \frac{e^\phi}{(1 + e^\phi)^2} = p_j (1 - p_j)$$

$$\therefore l = n \sum_{j=1}^{k-1} [z_j \phi_j - z_{j+1} g(\phi_j)]$$

$$u_j = \frac{\partial l}{\partial \phi_j} = n [z_j - z_{j+1} g'(\phi_j)] \quad \text{--- (1)}$$

$$\Rightarrow u_j = n \left[z_j - z_{j+1} \frac{y_j}{y_{j+1}} \right]$$

For max likelihood,

$$z_j - z_{j+1} \frac{y_j}{y_{j+1}} = 0$$

$$\Rightarrow \frac{z_j}{z_{j+1}} = \frac{y_j}{y_{j+1}} = \frac{z_j}{z_{j+1}}$$

$\Rightarrow$ estimated conditional probability = observed conditional proportion

$$\frac{\partial^2 l}{\partial \phi_j^2} = -n z_{j+1} g''(\phi_j) \quad [\text{From (1)}]$$

$$I_j = -n z_{j+1} p_j (1-p_j)$$

$\therefore$ Fisher information,

$$I_j = n z_{j+1} p_j (1-p_j)$$

The model parameters are:

$$\gamma = (\theta_1, \theta_2, \dots, \theta_{k-1}, \beta_1, \dots, \beta_p)^T$$

ϕ_j depends on these parameters

$$\Rightarrow \phi_j = \phi_j(\gamma)$$

$$\Rightarrow \frac{\partial l}{\partial \gamma} = \sum_{j=1}^{k-1} \frac{\partial l}{\partial \phi_j} \frac{\partial \phi_j}{\partial \gamma}$$

Let $u(\gamma) = \frac{\partial l}{\partial \gamma}$ be score vector

and

$$A(\gamma) = -E \left[\frac{\partial^2 \ell}{\partial \gamma \partial \gamma^T} \right] \quad \text{be fisher information matrix}$$

At MLE, $u(\gamma) = 0$

$\Rightarrow$ expanding the score around current estimate $\gamma^{(t)}$,

$$u(\gamma^{(t+1)}) \approx u(\gamma^{(t)}) + \underbrace{\frac{\partial u}{\partial \gamma^T}}_{= -A(\gamma)} (\gamma^{(t+1)} - \gamma^{(t)})$$

$$\Rightarrow u^{(t)} = A^{(t)} [\gamma^{(t+1)} - \gamma^{(t)}]$$

$$\gamma^{(t+1)} = \gamma^{(t)} + [A^{(t)}]^{-1} u^{(t)}$$

This is the fisher scoring update

This is repeated until convergence. McCullagh reports that roughly 4-5 cycles were enough for 4 significant digit accuracy.

4.

(a)

For one observation (x_n, t_n)

$$t_n | x_n, w \sim \mathcal{N}(w^T \phi(x_n), \sigma_n^2)$$

$$\Rightarrow p(t_n | x_n, w) = \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp \left[-\frac{(t_n - w^T \phi(x_n))^2}{2\sigma_n^2} \right]$$

PRIOR:

$$p(w|\alpha) = \mathcal{N}(w|0, \alpha^{-1}I)$$

$$= \left(\frac{\alpha}{2\pi} \right)^{M/2} \exp \left[-\frac{\alpha}{2} w^T w \right]$$

$$\therefore p(t_n | x_n, w) = \mathcal{N}(t_n | w^T \phi(x_n), \sigma_n^2)$$

$$\text{and } p(w|\alpha) = \mathcal{N}(w|0, \alpha^{-1}I)$$

(b)

~~Let~~

$$\text{Let } D = \{(x_n, t_n)\}_{n=1}^N$$

Assumptions: ~~are~~ observations are iid

$$p(t | X, w) = \prod_{n=1}^N p(t_n | x_n, w)$$

$$= \prod_{n=1}^N \frac{1}{\sqrt{2\pi\sigma_n^2}} \exp \left[-\frac{(t_n - w^T \phi(x_n))^2}{2\sigma_n^2} \right]$$

Taking ~~the~~ log,

$$= -\frac{1}{2} \sum_{n=1}^N \left[\log(2\pi\sigma_n^2) + \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} \right]$$

$$w_{MLE} = \arg \max_w \left[-\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} \right]$$

$$= \arg \min_w \left[\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} \right]$$

For MAP,

$$p(w|x, t) \propto p(t|x, w) p(w)$$

$$\Rightarrow \log p(w|x, t) = \log p(t|x, w) + \log p(w) + \text{constant}$$

$$\text{WKT//} \log p(w) = \frac{\alpha}{2} w^T w + \text{constant}$$

$$\begin{aligned} \therefore w_{\text{MAP}} &= \arg \max_w \left[-\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} - \frac{\alpha}{2} w^T w \right] \\ &= \arg \min_w \left[\frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2} + \frac{\alpha}{2} w^T w \right] // \end{aligned}$$

(c) Conv

$$E_D(w) = \frac{1}{2} \sum_{n=1}^N r_n (t_n - w^T \phi(x_n))^2$$

$$\text{WKT//} = \frac{1}{2} \sum_{n=1}^N \frac{(t_n - w^T \phi(x_n))^2}{\sigma_n^2}$$

$$\Rightarrow r_n = \frac{1}{\sigma_n^2}$$

Taking derivative, & equating to 0

$$0 = -\frac{1}{2} \sum_{n=1}^N r_n \phi_n (t_n - w^T \phi_n) [\phi_n = \phi(x_n)]$$

$$\Rightarrow \sum_{n=1}^N r_n \phi_n t_n = \sum_{n=1}^N r_n \phi_n \phi_n^T w$$

$$\Rightarrow w_{\text{ML}} = \left[\sum_{n=1}^N r_n \phi(x_n) \phi(x_n)^T \right]^{-1} \left[\sum_{n=1}^N r_n \phi(x_n) t_n \right]$$

$$\text{Let } R = \begin{bmatrix} r_1 & 0 & \dots & 0 \\ 0 & r_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_N \end{bmatrix} \quad t = \begin{bmatrix} t_1 \\ \vdots \\ t_N \end{bmatrix} \quad \Phi = [\phi(x_1) \dots \phi(x_N)]$$

$$\Rightarrow \text{Objective: } = \frac{1}{2} (t - \Phi^T w)^T R (t - \Phi^T w)$$

Differentiating & equating to zero,

$$\phi^T R (t - \phi^T w) = 0$$

$$\Rightarrow \phi^T R t = \phi^T R \phi^T w$$

$$\Rightarrow w_{ML} = (\phi^T R \phi^T)^{-1} \phi^T R t //$$

5.

(a) In logistic regression, for each data point,

$$t_n \in \{0, 1\}$$

WKT//

$$p(t_n = 1 | x_n, w) = \sigma(w^T x_n) = \frac{1}{1 + e^{-w^T x_n}}$$

$$\text{Let } y_n = \sigma(w^T x_n)$$

$$\Rightarrow p(t_n = 1 | x_n, w) = y_n$$

$$\therefore p(t_n = 0 | x_n, w) = 1 - y_n //$$

For a pt,

$$p(t_n | x_n, w) = y_n^{t_n} (1 - y_n)^{1 - t_n}$$

For N observations,

$$p(t | X, w) = \prod_{n=1}^N y_n^{t_n} (1 - y_n)^{1 - t_n}$$

Taking log.

$$= \sum_{n=1}^N [t_n \log y_n + (1 - t_n) \log (1 - y_n)]$$

$$E(w) = - \sum_{n=1}^N [t_n \log y_n + (1 - t_n) \log (1 - y_n)]$$

Taking the gradient,

$$\begin{aligned} \nabla E(w) &= \sum_{n=1}^N (y_n - t_n) x_n \\ &= \sum_{n=1}^N [\sigma(w^T x_n) - t_n] x_n // \end{aligned}$$

$$\text{Let } \phi = [x_1 \quad x_2 \quad \dots \quad x_N], \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{bmatrix} \quad \& \quad t = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix}$$

$$\Rightarrow \nabla E(w) = \phi(y - t)$$

Differentiating gradient again,

at each point,

$$\frac{\partial y_n}{\partial w} = y_n(1-y_n)x_n$$

$$\text{Let } r_n = y_n(1-y_n)$$

$$\& \quad R = \begin{bmatrix} r_1 & 0 & \dots & 0 \\ 0 & r_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & r_N \end{bmatrix}$$

$$\therefore \text{Hessian, } H = \nabla^2 E(w) = \Phi R \Phi^T$$

Newton-Raphson method to minimize $E(w)$:

$$w_{\text{new}} = w_{\text{old}} - H^{-1} \nabla E$$

Substituting the values,

$$w_{\text{new}} = w_{\text{old}} - (\Phi R \Phi^T)^{-1} \Phi (y - t)$$

$$w_{\text{new}} = w_{\text{old}} + (\Phi R \Phi^T)^{-1} \Phi (t - y) //$$

Algorithm

Step 1: Initialize : Choose $w^{(0)}$

Step 2: Calculate predictions, for each n ,

$$y_n^{(k)} = \sigma((w^{(k)})^T x_n)$$

Step 3: Construct R

$$r_n^{(k)} = y_n^{(k)}(1 - y_n^{(k)})$$

$$R^{(k)} = \text{diag}(r_1^{(k)}, r_2^{(k)}, \dots, r_N^{(k)})$$

Step 4: Calculate Gradient:

$$\nabla E(w^{(k)}) = \Phi (y^{(k)} - t)$$

Step 5: Calculate Hessian

$$H^{(k)} = \Phi R^{(k)} \Phi^T$$

Step 6: Update weights

$$\begin{aligned} \omega^{(k+1)} &= \omega^{(k)} - (H^{(k)})^{-1} \nabla E(\omega^{(k)}) \\ &= \omega^{(k)} + (\Phi R^{(k)} \Phi^T)^{-1} \Phi (t - y^{(k)}) \end{aligned}$$

Step 7: Repeat until convergence (change in ω or $E(\omega) < \text{threshold}$)

Step 8: Return ω

(b) WKTH

$$\begin{aligned} \omega_{\text{new}} &= \omega_{\text{old}} + (\Phi R \Phi^T)^{-1} \Phi^T (t - y) \\ &= (\Phi R \Phi^T)^{-1} [\omega (\Phi R \Phi^T) \omega_{\text{old}} + \Phi^T (t - y)] \\ &= (\Phi R \Phi^T)^{-1} \Phi R [\Phi^T \omega_{\text{old}} + R^{-1} (t - y)] \end{aligned}$$

$$\text{Let } z = \Phi^T \omega_{\text{old}} + R^{-1} (t - y)$$

$$\Rightarrow \omega_{\text{new}} = (\Phi R \Phi^T)^{-1} \Phi R z$$

which is exactly the weighted least squares solution from 4(c) ~~was~~ except weights $r_n = y_n \cdot (1 - y_n)$ and target t_n replaced by pseudo targets z_n

Unlike the std weighted least squares, the matrix R & effective target z both depend on ω_{old} through predictions y .

$\therefore R$ & z must be re-evaluated at every iteration based on ω_{new} , the optimization technique is named Iteratively Reweighted Least Squares

(c)

WK7//

$$H = \Phi R \Phi^T \quad \text{where } r_n = y_n(1-y_n)$$

$$\because 0 < y_n < 1 \Rightarrow y_n(1-y_n) > 0$$

$$\therefore R \geq 0 \quad [R \text{ is +ve semi definite}]$$

Consider an arbitrary vector a ,

$$\begin{aligned} a^T H a &= a^T \Phi R \Phi^T a \\ &= (\Phi^T a)^T R (\Phi^T a) \end{aligned}$$

$\because R$ is positive semi-definite,

$$(\Phi^T a)^T R (\Phi^T a) \geq 0$$

$$\therefore H \geq 0$$

$\therefore E(w)$ is convex

If H is positive definite, then $E(w)$ is strictly convex

$\Rightarrow$ there can only be one global minimum

$$\therefore H = \Phi R \Phi^T$$

$\Rightarrow H$ is positive definite when Φ has full row rank

$\therefore$ if Φ has full row rank

$$\Rightarrow H \succ 0$$

$\Rightarrow E(w)$ is strictly convex

$\Rightarrow$ unique global minimum.